SYSTEMATIC SEARCH FOR SINGULARITIES IN 3D EULER-VOIGT FLOWS

SYSTEMATIC SEARCH FOR SINGULARITIES IN 3D EULER-VOIGT FLOWS

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Lay Abstract

The Navier-Stokes equations are a set of equations used to model the behaviour of viscous incompressible fluids. The question of whether or not the Navier-Stokes equations, or the inviscid Euler equations, are a valid description of fluid mechanics remains unanswered. A singularity is the occurrence of an infinite derivative of the velocity at a single point in the fluid flow. If the equations form a singularity from smooth initial conditions, then they are not valid descriptions of fluid mechanics. The Euler-Voigt equations are an inviscid regularization of the Euler equations which are known to not form singularities from smooth initial conditions. However, by decreasing the regularization parameter we approximate Euler flows. We examine the Euler-Voigt equations with a PDE-constrained optimization problem to identify initial conditions that produce 'extreme' behaviour in numerical simulations. ## We present numerical evidence that CONCLUSION ##

Abstract

An abstract of not more than 300 words must be included and indicates the major emphasis of the thesis, new discoveries, and its contribution to knowledge. The style of abstract varies from discipline to discipline but the student should follow an appropriate style. This page must be numbered 'iv'.

The Navier-Stokes equations are a set of partial differential equations used to model the behaviour of viscous incompressible fluids. The question of whether or not unique classical solutions of the Navier-Stokes equations, and inviscid Euler equations, exist for all time, given smooth initial conditions, remains unanswered for the three-dimensional problem.

A singularity is the occurrence of an infinite derivative of the velocity at a single point in the fluid flow. The formation of a singularity from smooth initial conditions would invalidate the equations as accurate descriptions of fluid mechanics. The Euler-Voigt equations are an inviscid regularization of the Euler equations which are known to not form singularities from smooth initial conditions. However, by decreasing the regularization parameter we approximate Euler flows. We examine the Euler-Voigt equations with a PDE-constrained optimization problem to identify initial conditions that produce 'extreme' behaviour in numerical simulations. ## We present numerical evidence that CONCLUSION ##

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An expression of thanks for assistance given by the Supervisor of research and by others should be either set forth on a separate page or incorporated into the Preface (if there is one). These and all subsequent preliminary pages listed under (f), (g) and (h) must be numbered in lower case Roman numerals, i.e., ‘iv’, ‘v’, ‘vi’ etc.

Dr. Bartosz Protas, Dr. Xinyu Zhao, Dr. Jörn Davidsen, members of committee, Parents

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Partial Differential Equation - PDE

Declaration of Academic Achievement

The student will declare his/her research contribution and, as appropriate, those of colleagues or other contributors to the contents of the thesis.

1 Introduction

1.1 Singularity Formation in 3D Fluid Flows

The Navier-Stokes equations are a set of partial differential equations (PDEs) that describe the motion of viscous incompressible fluids. The fluids described by the Navier-Stokes equations can be modelled in domains with a wide variety of shape, dimension, and boundary conditions [CITATION]. Because of their flexibility they are utilized to model the behaviour of systems in fields as disparate as aero/hydrodynamics, weather prediction, biophysics, and astrophysics [2][3]. Despite the numerous fields that make use of the 3D Navier-Stokes equations, and extensive research examining their properties, there is no known unique classical solution [7]. In fact, this is the basis of the well known millennium problem from the Clay Institute which offers a million dollar award for the confirmation of the existence, or non-existence, of a solution for smooth initial conditions [8].

Most often, scientific research that focuses the fundamental properties of the Navier-Stokes equations utilizes either the unbounded domain $\Omega := \mathbb{R}^3$, or the three-dimensional torus $\Omega := \mathbb{T}^3_L$ with $L > 0$ [7]. For this analysis, we will use the three-dimensional torus. The Navier-Stokes system on the domain $\Omega := \mathbb{T}^3_L := [0, L]^3$ and time interval $[0, T]$ is defined as

$$\partial_t \boldsymbol{u} + \boldsymbol{u} \cdot \boldsymbol{\nabla} \boldsymbol{u} + \boldsymbol{\nabla} p - \nu \Delta \boldsymbol{u} = 0 \qquad \text{in } \Omega \times (0, T], \tag{1.1a}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{u} = 0 \qquad \text{in } \Omega \times [0, T], \tag{1.1b}$$

$$\boldsymbol{u}(0) = \boldsymbol{\eta} \tag{1.1c}$$

where $\boldsymbol{u} = [u_1, u_2, u_3]^T$ is the fluid velocity field, p is the scalar pressure, and $\nu > 0$ is the coefficient of kinematic viscosity. The value $\boldsymbol{\eta}$ is the initial divergence free velocity field of the fluid at time $t = 0$. Finally, the $\boldsymbol{\nabla} = [\partial_1, \partial_2, \partial_3]^T$ operator is used to define the gradient of the pressure, the tensor $[\boldsymbol{\nabla}\boldsymbol{u}]_{ij} = \partial_j u_i$ for $i, j = 1, 2, 3$, and the divergence of the velocity field.

One line of evidence against the validity of the 3D Navier-Stokes equations is the formation of singularities. A singularity occurs when the derivative of the velocity field becomes infinite at a single point [6]. *How do they form?*

The formation of a singularity in a fluid flow given smooth initial conditions would invalidate the Navier-Stokes equations as an accurate description of fluid flows [7]. The question of singularity formation from smooth initial conditions has also not been answered for the Euler equations. The Euler equations are the inviscid form of the Navier-Stokes equations, obtained by setting $\nu = 0$

$$\partial_t \boldsymbol{u} + \boldsymbol{u} \cdot \boldsymbol{\nabla}\boldsymbol{u} + \boldsymbol{\nabla} p = 0 \qquad \text{in } \Omega \times (0, T], \tag{1.2a}$$

$$\boldsymbol{\nabla} \cdot \mathbf{u} = 0 \qquad \text{in } \Omega \times [0, T], \tag{1.2b}$$

$$\boldsymbol{u}(0) = \boldsymbol{\eta} \tag{1.2c}$$

The Euler are often studied because they are simpler to analyze than the Navier-Stokes equations [Citation]. Additionally, many people believe the Eulers equations are more likely to form singularities than the Navier-Stokes due to the lack of a viscosity term dampening the fluid flow [Citation].

1.2 3D Euler-Voigt Equations

The Euler-Voigt equations are an inviscid regularization of the Euler equations (1.2) which have been shown to be globally well-posed and so do not form singularities given smooth initial conditions [1]. The Euler-Voigt equations on the torus $\Omega = \mathbb{T}_L^3 := [0, L]^3$ and time interval $[0, T]$ are defined as

$$-\alpha^2 \partial_t \Delta \boldsymbol{u}_\alpha + \partial_t \boldsymbol{u}_\alpha + \boldsymbol{u}_\alpha \cdot \boldsymbol{\nabla} \boldsymbol{u}_\alpha + \boldsymbol{\nabla} p = 0 \qquad \text{in } \Omega \times (0, T], \tag{1.3a}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{u}_\alpha = 0 \qquad \text{in } \Omega \times (0, T], \tag{1.3b}$$

$$\boldsymbol{u}_\alpha(x, 0) = \boldsymbol{\eta}_\alpha. \tag{1.3c}$$

where $\boldsymbol{u}_\alpha$ is the velocity field of the Euler-Voigt flow. The p and $\boldsymbol{\nabla}$ symbols have the same definitions as systems (1.1) and (1.2). The regularization of the Euler-Voigt equations is accomplished with the addition of the term $-\alpha^2 \partial_t \Delta \boldsymbol{u}_\alpha$ where $\alpha > 0$ is the regularization parameter. This term is used to stabilize simulations without adding artificial viscosity, which would remove energy from the system [4]. The property of constant energy, known as the 'α-energy' can be used to test the validity of the simulations. The α-energy is defined as

$$E_\alpha(t) := \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 = E_\alpha(0). \tag{1.4}$$

It is worth noting that both (1.2) and (1.3) possess the following time scaling symmetry

$$\tilde{\boldsymbol{u}}(\boldsymbol{x}, t) = \lambda \boldsymbol{u}(\boldsymbol{x}, \lambda t) \tag{1.5}$$

Why study the Euler-Voigt equations.

We study the Euler-Voigt equations because they are faster and let us explore the euler-voigt equations. By modelling the Euler-Voigt flows at decreasing values of α we approximate Euler flows. Based on prior communication with Dr. Xinyu Zhao. The behavior of the solutions of (1.3) as $\alpha \to 0$ can be used to determine whether the solutions of the Euler equations will develop a finite time singularity [4]

1.3 Initial Conditions

The specific initial conditions that will be used in our simulations are described in section NUMBER. However, in order to formulate our optimization problem we must first describe the properties that each initial condition must have. Firstly, the pseudospectral methods used in this study require the functions to be smooth and real-analytic in order to have exponential convergence. A requirement of the Euler-Voigt equations is that the initial conditions are divergence free. Additionally, we assume without loss of generality that the initial conditions have a mean of zero. *This is important because REASON.*

In order to identify initial conditions that satisfy these conditions, we consider analytic initial conditions that belong to the following Gevrey class

$$G^\sigma := \left\{ \boldsymbol{v} \in H^s(\mathbb{T}^3) : \|\boldsymbol{v}\|_{G^\sigma} = \langle \boldsymbol{v}, \boldsymbol{v} \rangle_{G^\sigma}^{1/2} < \infty \right\}, s \geq 3, \sigma > 0, \tag{1.6}$$

with the inner product

$$\forall \boldsymbol{v}, \boldsymbol{u} \in G^\sigma, \quad \langle \boldsymbol{v}, \boldsymbol{u} \rangle_{G^\sigma} := \sum_{\boldsymbol{j} \in \mathbb{Z}^3} (1 + |2\pi \boldsymbol{j}|^2)^s e^{4\pi\sigma|\boldsymbol{j}|} \hat{\boldsymbol{v}}_{\boldsymbol{j}} \cdot \overline{\hat{\boldsymbol{u}}}_{\boldsymbol{j}} \tag{1.7a}$$

$$= \int_{\mathbb{T}^3} (1 + |D|^2)^s e^{2\sigma|D|} \boldsymbol{v} \cdot \boldsymbol{u} \, dx, \tag{1.7b}$$

where overbar denotes complex conjugation, whereas the operators $|D|$ and $e^{\sigma|D|}$ are defined via

$$\left[\widehat{|D|\boldsymbol{v}}\right]_{\boldsymbol{j}} := 2\pi|\boldsymbol{j}|\hat{\boldsymbol{v}}_{\boldsymbol{j}}, \qquad \left[\widehat{e^{\sigma|D|}\boldsymbol{v}}\right]_{\boldsymbol{j}} := e^{2\pi|\boldsymbol{j}|}\hat{\boldsymbol{v}}_{\boldsymbol{j}}. \tag{1.8}$$

Since the initial conditions of (1.3) need to be divergence free and the mean is an invariant of motion, we introduce the following subspace in which optimal initial conditions will be sought

$$V := \{\boldsymbol{v} \in G^{\sigma} : \boldsymbol{\nabla} \cdot \boldsymbol{v} = 0, \int_{\mathbb{T}^3} \boldsymbol{v}\, dx = 0\}. \tag{1.9}$$

Due to the time scaling property of the Euler-Voigt equations (1.5), we can restrict our discussion to initial conditions with unit $\dot{H}^1$ seminorm which belong to a closed manifold $\mathcal{M}_1 \subset V$ defined as

$$\mathcal{M}_1 = \{\boldsymbol{v} \in V : \|\boldsymbol{v}\|_{\dot{H}^1} = 1\}. \tag{1.10}$$

The $\dot{H}^1$ seminorm is defined as

$$\|v\|_{\dot{H}^1} = (2\pi)^3 \sum_{k \in \dot{\mathbb{Z}}^3} |k|^2 |\hat{\boldsymbol{u}}_k|^2 \tag{1.11}$$

where

$$\dot{Z Z}^3 := \left\{k \in \mathbb{Z}^3 : |k| \neq 0\right\}.$$

1.4 Finite Time Blow-up Criteria

As previously mentioned, the behaviour of the Euler-Voigt equations as $\alpha \to 0^+$ can be used to to determine whether solutions of the Euler equations will develop a finite time singularity. This was proved in the following theorem from [4].

Theorem 1.1. *Let $\eta_\alpha \in H^s$,$s \geq 3$ with $\nabla \cdot \eta_\alpha = 0$ and let $\boldsymbol{u}_\alpha$ be the corresponding unique solution of the Euler-Voigt equations. Suppose that there exists a time $T^* > 0$ such that*

$$\limsup_{\alpha \to 0^+} \left(\alpha \sup_{t \in [0,T^*]} \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2} \right) > 0. \tag{1.12}$$

Then the Euler equations with the initial condition $\boldsymbol{\eta}_\alpha$ must develop a singularity within the *interval* $[0, T^*]$.

Theorem 5.1 in [5] states that

Theorem 1.2. *Given initial data* $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ *for some* $s \geq 3$*, let* $\boldsymbol{u}, \boldsymbol{u}_\alpha \in C([0,T], H^s(\mathbb{T}^3)) \cap C^1([0,T], H^{s-1}(\mathbb{T}^3))$ *be the corresponding solutions to (1.2) and (1.3), respectively, where* $0 < T < T_{max}$ *and* T_{max} *is the maximal time for which a solution to (1.2) exists and is unique. For all* $t \in [0,T]$*, we have*

$$\begin{aligned} &\|\boldsymbol{u}(t) - \boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{u}(t) - \boldsymbol{u}_\alpha(t))\|_{L^2}^2 \leq \\ &\left(\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2\right) e^{Ct} + C\alpha^2(e^{Ct} - 1). \end{aligned} \tag{1.13}$$

Based on this theorem we know that if

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \to 0, \quad \text{as } \alpha \to 0, \tag{1.14}$$

then for all $t \in [0,T]$,

$$\|\boldsymbol{u}(t) - \boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{u}(t) - \boldsymbol{u}_\alpha(t))\|_{L^2}^2 \to 0. \tag{1.15}$$

Finally, we slightly modify Theorem 5.2 in [5] as follows.

Theorem 1.3. *Given initial data* $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ *for some* $s \geq 3$ *satisfying*

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \to 0, \quad as \alpha \to 0, \tag{1.16}$$

let $\boldsymbol{u}, \boldsymbol{u}_\alpha \in C([0,T], H^s(\mathbb{T}^3)) \cap C^1([0,T], H^{s-1}(\mathbb{T}^3))$ *be the corresponding solutions to (1.2) and (1.3), respectively. If there exists a finite time* $T^* > 0$ *such that the solutions* $\boldsymbol{u}_\alpha$ *satisfies*

$$\limsup_{\alpha^2 \to 0^+} \left(\alpha \sup_{t \in [0,T^*]} \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right) > 0, \tag{1.17}$$

then $\boldsymbol{u}$ develops a singularity in the interval $[0, T^*]$.

Proof. Suppose that the solution $\boldsymbol{u}$ of the Euler equations (1.2) with the initial condition $\boldsymbol{\eta}$ is regular on $[0, T^*]$. Since the Euler-Voigt equations are globally well-posed, we have

$$\|\boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2\|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2\|\boldsymbol{\nabla}(\boldsymbol{\eta}_\alpha)\|_{L^2}^2, \tag{1.18}$$

thus

$$\alpha^2\|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2\|\boldsymbol{\nabla}(\boldsymbol{\eta}_\alpha)\|_{L^2}^2 - \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2. \tag{1.19}$$

Taking the supremum over $t \in [0, T^*]$ of the above equation, we obtain that

$$\sup_{t\in[0,T^*]} \alpha^2\|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2\|\boldsymbol{\nabla}(\boldsymbol{\eta}_\alpha)\|_{L^2}^2 - \inf_{t\in[0,T^*]} \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2. \tag{1.20}$$

Taking the limsup when $\alpha \to 0$ of the above equation, we obtain that

$$\limsup_{\alpha\to 0}\left(\sup_{t\in[0,T^*]} \alpha^2\|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2\right) = \limsup_{\alpha\to 0}\left(\|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2\|\boldsymbol{\nabla}(\boldsymbol{\eta}_\alpha)\|_{L^2}^2 - \inf_{t\in[0,T^*]} \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2\right). \tag{1.21}$$

We now make use of Theorem (1.2), which demonstrates that the convergence of the solutions of the Euler-Voigt equations (1.3) to the solution of the Euler equations (1.2) is uniform on $[0, T^*]$. Therefore we can reduce the above equation to

$$\limsup_{\alpha\to 0}\left(\sup_{t\in[0,T^*]} \alpha^2\|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2\right) = \|\boldsymbol{\eta}\|_{L^2}^2 - \|\boldsymbol{u}(t)\|_{L^2}^2. \tag{1.22}$$

Since the energy of regular solutions of the Euler equations is conserved, we have

$$\limsup_{\alpha\to 0}\left(\sup_{t\in[0,T^*]} \alpha^2\|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2\right) = 0, \tag{1.23}$$

which contradicts the hypothesis in the theorem that the above expression is positive. □

2 The Optimization Problem

2.1 Formulation of the Problem

With our initial conditions and blowup criteria defined, we can now define the objection function $\Phi_{\alpha,T} : V \to \mathbb{R}^+$ as

$$\Phi_{\alpha,T}(\boldsymbol{\eta}) := \|\nabla \boldsymbol{u}_\alpha(T;\boldsymbol{\eta})\|_{L^2}^2 = \|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta})\|_{\dot{H}^1}^2, \tag{2.1}$$

where $\boldsymbol{u}_\alpha(T;\boldsymbol{\eta})$ is the solution of the Euler-Voigt equations obtained with the initial condition $\boldsymbol{\eta} \in V$. We can now formulate our optimization problem. Given $\alpha, T \in \mathbb{R}_+$, find

$$\tilde{\boldsymbol{\eta}}_{\alpha,T} = \underset{\boldsymbol{\eta} \in M_1}{\operatorname{argmax}}\, \Phi_{\alpha,T}(\boldsymbol{\eta}). \tag{2.2}$$

The objective function is maximized using an iterative relation

$$\boldsymbol{\eta}_{\alpha,T}^{n+1} = \boldsymbol{\eta}_{\alpha,T}^{n} + \tau_n \nabla\Phi_T\left(\boldsymbol{\eta}_{\alpha,T}^{n}\right), \quad \boldsymbol{\eta}_{\alpha,T}^{(0)} = \boldsymbol{\eta}_{\alpha,0}, \quad n = 0, 1, ... \tag{2.3}$$

where $\nabla\Phi_T$ is the gradient of the objective functional evaluated at the element η_T^n, τ_n is the step size along the gradient direction, and η_0 is the first initial condition. The solution to the problem is the limit as $n \to \infty$. Our goal is to first fix T and solve the optimization problem for a sequence of decrease values of α that converge to 0.

2.2 Gâteaux (Directional) Differential

We compute the gradient $\nabla\Phi_{\alpha,T}(\boldsymbol{\eta})$ of the objection functional using an adjoint method which depends on solving a properly defined adjoint system backwards in time. We first introduce the Gâteaux (directional) differential $\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') : V \times V \to \mathbb{R}$. The equation for the Gateaux differential is,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') := \lim_{\epsilon\to 0}\frac{1}{\epsilon}\Phi_{\alpha,T}(\boldsymbol{\eta}+\epsilon\boldsymbol{\eta}') - \Phi_{\alpha,T}(\boldsymbol{\eta})]. \tag{2.4}$$

Substituting the second definition of the objective functional (2.1) into the Gâteaux differential, we derive the Gâteaux differential in the form of the $\dot{H}^1$ inner product and the L^2 inner product,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle \boldsymbol{u}_\alpha(\cdot,T), \boldsymbol{u}'_\alpha(\cdot,T)\right\rangle_{\dot{H}^1} = 2\left\langle \boldsymbol{u}^*_\alpha(\cdot,0), \boldsymbol{u}'_\alpha(\cdot,0)\right\rangle_{L^2} \tag{2.5}$$

The complete derivations for both forms are shown in Appendix 6.1. In (2.5) the value $\boldsymbol{u}'_\alpha$ is the solution of the linearization of the Euler-Voigt system (1.3) around its solution $\boldsymbol{u}_\alpha$. The linearization of the Euler-Voigt system is defined as

$$\begin{aligned}\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix} &:= \begin{bmatrix}\partial_t\boldsymbol{u}'_\alpha + (I-\alpha^2\Delta)^{-1}(\boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \boldsymbol{\nabla}p') = 0\\ \nabla\cdot\boldsymbol{u}'_\alpha = 0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix} \\ \boldsymbol{u}'_\alpha(x,0) &= \boldsymbol{\eta}'\end{aligned} \tag{2.6}$$

where $\boldsymbol{\eta}'$ and p' are the perturbtionations of the initial condition and pressure, respectively. The value $\boldsymbol{u}_\alpha^*$ is the solution of the adjoint system

$$\mathcal{L}^* \begin{bmatrix} \boldsymbol{u}_\alpha^* \\ p_\alpha^* \end{bmatrix} := \begin{bmatrix} -\partial_t \boldsymbol{u}_\alpha^* - (\boldsymbol{\nabla}(I-\alpha^2)^{-1}\boldsymbol{u}_\alpha^* + (\boldsymbol{\nabla}(I-\alpha^2)^{-1}\boldsymbol{u}_\alpha^*)^T)\boldsymbol{u}_\alpha - \boldsymbol{\nabla} p_\alpha^* \\ -\boldsymbol{\nabla}\cdot\boldsymbol{u}_\alpha^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{2.7}$$
$$\boldsymbol{u}_\alpha^*(x,T) = 2|D|^2 \boldsymbol{u}_\alpha(\boldsymbol{x}.T),'$$

which is derived by performing integration by parts, in both space and time, on the following duality-pairing relation (6.3)

$$\begin{aligned} \left(\mathcal{L}\begin{bmatrix} \boldsymbol{u}_\alpha' \\ p' \end{bmatrix}, \begin{bmatrix} \boldsymbol{u}_\alpha^* \\ p^* \end{bmatrix}\right) &:= \int_0^T \int_{\mathbb{T}^3} \mathcal{L}\begin{bmatrix} \boldsymbol{u}_\alpha' \\ p' \end{bmatrix}, \begin{bmatrix} \boldsymbol{u}_\alpha^* \\ p^* \end{bmatrix} dx\,dt \\ &= \left(\begin{bmatrix} \boldsymbol{u}_\alpha' \\ p' \end{bmatrix}, \mathcal{L}^*\begin{bmatrix} \boldsymbol{u}_\alpha^* \\ p^* \end{bmatrix}\right) + \int_{\mathbb{T}^3} \boldsymbol{u}_\alpha'(\boldsymbol{x},T)\cdot\boldsymbol{u}_\alpha^*(\boldsymbol{x},T)\,dx \\ -\int_{\mathbb{T}^3} \boldsymbol{\eta}'(\boldsymbol{x})\cdot\boldsymbol{u}_\alpha^*(\boldsymbol{x},0)\,dx \\ &= 0. \end{aligned} \tag{2.8}$$

Using the adjoint system we can now define the gradient. We take the Gâteaux differential (2.5) and the Riesz representation form

$$\Phi_{\alpha,T}'(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle \boldsymbol{u}_\alpha^*(\cdot,0), \boldsymbol{u}_\alpha'(\cdot,0)\right\rangle_{L^2} = \left\langle \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}), \boldsymbol{\eta}'\right\rangle. \tag{2.9}$$

Expressing the L^2 inner product in its integral form,

$$\langle \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}), \boldsymbol{\eta}' \rangle = \int_{\mathbb{T}^3} \boldsymbol{u}_\alpha^*(x,0) \cdot \boldsymbol{\eta}'(x)\, dx \tag{2.10}$$

$$= \int_{\mathbb{T}^3} (1+|D|^2)^3 e^{2\sigma|D|} \left[(1+|D|^2)^{-3} e^{-2\sigma|D|} \boldsymbol{u}_\alpha^*(x,0)\right] \cdot \boldsymbol{\eta}'(x)\, dx \tag{2.11}$$

$$\left\langle (1+|D|^2)^{-3} e^{-2\sigma|D|} \boldsymbol{u}_\alpha^*(x,0), \boldsymbol{\eta}'(x) \right\rangle_{G^\sigma} \tag{2.12}$$

Therefore, the gradient of the objective functional is

$$\nabla\Phi_{\alpha,T}(\boldsymbol{\eta}) = (1+|D|^2)^{-3} e^{-2\sigma|D|} \boldsymbol{u}_\alpha^*(x,0)$$

2.3 Riemannian Conjugate Gradient Method

The optimization looks for an initial condition $\boldsymbol{\eta} \in M_1$ that maximizes our objective functional. Therefore, not only must our first initial condition $\boldsymbol{\eta}_{\alpha,T}^0$ have a unit $\dot{H}^1$ seminorm, but so to must each following iteration $\boldsymbol{\eta}_{\alpha,T}^{n+1}$. This will not be true in general given the iterative relation (2.3). We ensure this constraint is upheld using a Riemannian conjugate gradient method modified from the method used in [7]. This method has three steps:

1. Project the gradient $\nabla\Phi_T(\boldsymbol{\eta})$ onto the tangent space to $\mathcal{M}_1$ at $\boldsymbol{\eta}$
2. Perform a suitable vector transport operation in order to construct a Riemannian conjugate ascent direction using the previous search direction
3. Retract the resulting state back to the constraint manifold $\mathcal{M}_1$

3 Numerical Methods

3.1 Spatial Discretization & Spectral Methods

define the domain and mathematical tools used

3.2 Time Stepping

what time stepping methods are there. which time stepping method will we use.

3.3 Kappa Test Validation

The Euler-voigt equations (1.3) and adjoint system () are descritized in space using METHOD and time using METHOD. The kappa test is a means of validating these descritization methods based on the validation method of [**?**]. We define the value κ as

$$\kappa(\epsilon) = \frac{1}{\epsilon} \frac{[\Phi_{\alpha,T}(\boldsymbol{\eta} + \epsilon\boldsymbol{\eta}) - \Phi_{\alpha,T}(\boldsymbol{\eta})]}{\langle \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}), \boldsymbol{\eta}' \rangle} \tag{3.1}$$

where $\Phi_{\alpha,T}$ is the objective functional (2.1), $\boldsymbol{\eta}$ is the initial condition, and $\nabla\Phi_{\alpha,T}$ is the gradient of the objective functional (). The value η is the

4 Results

4.1 Results for $\alpha =$value 1

TEXT

4.2 Results for $\alpha =$value 2

TEXT

4.3 Results for α =value 3

TEXT

5 Conclusion

TEXT

6 Appendix

6.1 Derivation of the Gateaux Directional Differential

The equation for the Gateaux differential (2.4) is,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') := \lim_{\epsilon\to 0}\frac{1}{\epsilon}[\Phi_{\alpha,T}(\boldsymbol{\eta}+\epsilon\boldsymbol{\eta}') - \Phi_{\alpha,T}(\boldsymbol{\eta})].$$

The objective functional (2.1) is defined as,

$$\Phi_{\alpha,T}(\boldsymbol{\eta}) := \|\nabla\boldsymbol{u}_\alpha(T;\boldsymbol{\eta})\|^2_{L^2} = \|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta})\|^2_{\dot{H}^1}.$$

Substituting the second definition of (2.1) into (2.4),

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \lim_{\epsilon\to 0}\frac{1}{\epsilon}[\|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta}+\epsilon\boldsymbol{\eta}')\|^2_{\dot{H}^1} - \|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta})\|^2_{\dot{H}^1}].$$

The equation for the $\dot{H}^1$ seminorm is,

$$\|u\|_{\dot{H}^1} = (2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{\boldsymbol{u}}_k|^2.$$

Substituting in the equation for the $\dot{H}^1$ seminorm,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \lim_{\epsilon\to 0}\frac{1}{\epsilon}[(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{\boldsymbol{u}}_k+\epsilon\hat{\boldsymbol{u}}'_k|^2-(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{\boldsymbol{u}}_k|^2].$$

This simplifies to,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \lim_{\epsilon\to 0}\frac{1}{\epsilon}[(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2 2\epsilon|\hat{\boldsymbol{u}}_k\hat{\boldsymbol{u}}'_k|+(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2\epsilon^2|\hat{\boldsymbol{u}}'_k|^2].$$

Taking the limit $\epsilon\to 0$,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{\boldsymbol{u}}_k\hat{\boldsymbol{u}}'_k|.$$

This can be expressed in the integral form,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\int_{\mathbb{T}^3}\sum_{k\in\mathbb{Z}^3}|k|^2\hat{\boldsymbol{u}}_k\hat{\boldsymbol{u}}'_k e^{2ikx}dx.$$

Which is equivalent to the inner product,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle\sum_{k\in\mathbb{Z}^3}|k|\hat{\boldsymbol{u}}_k e^{ikx},\sum_{k\in\mathbb{Z}^3}|k|\hat{\boldsymbol{u}}'_k e^{ikx}\right\rangle$$

This gives us the Gateaux differential in the form of the $\dot{H}^1$ inner product,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle\boldsymbol{u}_\alpha(\cdot,T),\boldsymbol{u}'_\alpha(\cdot,T)\right\rangle_{\dot{H}^1}$$

The integral form of the $\dot{H}^1$ inner product is

$$\langle\boldsymbol{u},\boldsymbol{v}\rangle_{\dot{H}^1} = \int_{\mathbb{T}^3}|D|\boldsymbol{u}\cdot|D|\boldsymbol{v}dx.$$

where D is the operator defined in (1.8). Therefore, the Gateaux differential can be expressed as

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\int_{\mathbb{T}^3} |D|\boldsymbol{u}_\alpha(x,T)\cdot|D|\boldsymbol{u}'_\alpha(x,T)dx,$$

which can be rearranged to the form

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \int_{\mathbb{T}^3} 2|D|^2\boldsymbol{u}_\alpha(x,T)\cdot\boldsymbol{u}'_\alpha(x,T)dx.$$

We now make use of the adjoint system terminal condition derived in Appendix (6.3)

$$\boldsymbol{u}^*_\alpha(x,T) = 2|D|^2\boldsymbol{u}_\alpha(x,T).$$

Therefore

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \int_{\mathbb{T}^3} \boldsymbol{u}^*_\alpha(x,T)\cdot\boldsymbol{u}'_\alpha(x,T)dx.$$

Now using the duality-pairing defined in Appendix (6.3)

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \int_{\mathbb{T}^3} \boldsymbol{\eta}'(x)\cdot\boldsymbol{u}^*_\alpha(x,0)dx.$$

This is the inner product

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \left\langle \boldsymbol{u}^*_\alpha(x,0),\boldsymbol{\eta}'(x)\right\rangle_{L^2}.$$

Therefore,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle \boldsymbol{u}_\alpha(\cdot,T),\boldsymbol{u}'_\alpha(\cdot,T)\right\rangle_{\dot{H}^1} = \left\langle \boldsymbol{u}^*_\alpha(x,0),\boldsymbol{\eta}'(x)\right\rangle_{L^2}.$$

6.2 Linearization of the Euler-Voigt System

Euler-Voigt Equations,

$$\begin{cases} -\alpha^2\partial_t\Delta u_\alpha + \partial_t u_\alpha + (u_\alpha \cdot \nabla)u_\alpha + \nabla p = 0 \\ \nabla \cdot u_\alpha = 0 \\ u_\alpha|_{t=0} = \eta. \end{cases} \tag{6.1}$$

Substituting in $u_\alpha = u_\alpha + u'_\alpha$, $\eta = \eta + \eta'$, and $p = p + p'$,

$$\begin{cases} -\alpha^2\partial_t\Delta(u_\alpha + u'_\alpha) + \partial_t(u_\alpha + u'_\alpha) + ((u_\alpha + u'_\alpha) \cdot \nabla)(u_\alpha + u'_\alpha) + \nabla(p + p') = 0 \\ \nabla \cdot (u_\alpha + u'_\alpha) = 0 \\ (u_\alpha + u'_\alpha)|_{t=0} = \eta + \eta'. \end{cases}$$

Expanding,

$$\begin{cases} -\alpha^2\partial_t\Delta u_\alpha - \alpha^2\partial_t\Delta u'_\alpha + \partial_t u_\alpha + \partial_t u'_\alpha + u_\alpha \cdot \nabla u_\alpha + u_\alpha \cdot \nabla u'_\alpha + u'_\alpha \cdot \nabla u_\alpha + u'_\alpha \cdot \nabla u'_\alpha + \nabla p + \nabla p' = 0 \\ \nabla \cdot (u_\alpha + u'_\alpha) = 0 \\ (u_\alpha + u'_\alpha)|_{t=0} = \eta + \eta'. \end{cases}$$

Substituting in the equations from (3),

$$\begin{cases} -\alpha^2\partial_t\Delta u'_\alpha + \partial_t u'_\alpha + u_\alpha \cdot \nabla u'_\alpha + u'_\alpha \cdot \nabla u_\alpha + u'_\alpha \cdot \nabla u'_\alpha + \nabla p' = 0 \\ \nabla \cdot u'_\alpha = 0 \\ u'_\alpha|_{t=0} = \eta'. \end{cases}$$

Linearizing the system by removing non-linear terms, we derive the Linearized system,

$$\begin{cases} -\alpha^2 \partial_t \Delta u'_\alpha + \partial_t u'_\alpha + u_\alpha \cdot \nabla u'_\alpha + u'_\alpha \cdot \nabla u_\alpha + \nabla p' = 0 \\ \nabla \cdot u'_\alpha = 0 \\ u'_\alpha|_{t=0} = \eta'. \end{cases}$$

This can be rearranged to the form,

$$\begin{cases} \partial_t u'_\alpha + (I - \alpha^2 \Delta)^{-1}(u_\alpha \cdot \nabla u'_\alpha + u'_\alpha \cdot \nabla u_\alpha + \nabla p') = 0 \\ \nabla \cdot u'_\alpha = 0 \\ u'_\alpha|_{t=0} = \eta'. \end{cases} \tag{6.2}$$

6.3 Derivation of the Adjoint System

For this derivation we will exclude the α subscript on the u_α, u'_α, and u^*_α for the sake of simplicity. We derive the adjoint system using the following duality pairing relation,

$$(L[u'], u^*) := \int_0^T \int_{\mathbb{T}^3} L[u']u^* dxdt = (u', L[u^*]) + \int_a^b u'u^*|_{t=T} - \int_a^b u'u^*|_{t=0} dx = 0.$$

Substituting in the equation from the Euler-Voigt perturbation system,

$$(L[u'], u^*) := \int_0^T \int_{\mathbb{T}^3} (\partial_t u' + (I - \alpha^2 \Delta)^{-1}(u.\nabla u' + u' \cdot \nabla u_+ \nabla p'))u^* + (\nabla \cdot u')p^* dx\, dt.$$

This integral can be split into five parts.

Solving the first part with integration by parts, with $u = u^*$ and $v' = \partial_t u'$

$$\int_0^T \int_{\mathbb{T}^3} u^* \partial_t u' dx\, dt = \int_{\mathbb{T}^3} u^* u' |_{t=0}^{t=T} dx - \int_0^T \int_{\mathbb{T}^3} u' \partial_t u^* dx\, dt.$$

Starting with the second part and expressing it in Einstein Summation notation,

$$\int_0^T \int_{\mathbb{T}^3} \left[(I - \alpha^2 \Delta)^{-1} u \cdot \nabla u'\right] \cdot u^* dx\, dt = \int_0^T \int_{\mathbb{T}^3} \left[(I - \alpha^2 \Delta)^{-1} u_j \frac{\partial u_i'}{\partial x_j}\right] u_i^* dx\, dt.$$

Using the derivative product rule and the fact that $\nabla \cdot u = 0$,

$$\frac{\partial}{\partial x_j}(u_j u_i') = \frac{\partial u_j}{\partial x_j} u_i' + u_j \frac{\partial u_i'}{\partial x_j} = u_j \frac{\partial u_i'}{\partial x_j}.$$

Substituting this into the integral,

$$\int_0^T \int_{\mathbb{T}^3} \left[(I - \alpha^2 \Delta)^{-1} u \cdot \nabla u'\right] \cdot u^* dx\, dt = \int_0^T \int_{\mathbb{T}^3} \left[(I - \alpha^2 \Delta)^{-1} \frac{\partial}{\partial x_j}(u_j u_i')\right] u_i^* dx\, dt.$$

Using the periodic boundaries of the system, we can now apply integration by parts with $u = [(I - \alpha^2 \Delta)^{-1} u_i^*]$ and $v' = \frac{\partial}{\partial x_j}(u_j u_i')$,

$$\int_0^T \int_{\mathbb{T}^3} \left[(I - \alpha^2 \Delta)^{-1} u \cdot \nabla u'\right] \cdot u^* dx\, dt = -\int_0^T \int_{\mathbb{T}^3} u_j u_i' \frac{\partial}{\partial x_j} \left[(I - \alpha^2 \Delta)^{-1} u_i^*\right] dx\, dt.$$

Converting back to vector notation,

$$\int_0^T \int_{\mathbb{T}^3} \left[(I - \alpha^2 \Delta)^{-1} u \cdot \nabla u'\right] \cdot u^* dx\, dt = -\int_0^T \int_{\mathbb{T}^3} u' \cdot \left[u \cdot \nabla \left[(I - \alpha^2 \Delta)^{-1} u^*\right]\right] dx\, dt.$$

Starting with the third part and expressing it in Einstein Summation notation,

$$\int_0^T \int_{\mathbb{T}^3} \left[(I-\alpha^2\Delta)^{-1}u' \cdot \nabla u\right] \cdot u^* dx\, dt = \int_0^T \int_{\mathbb{T}^3} \left[(I-\alpha^2\Delta)^{-1}u'_j \frac{\partial u_i}{\partial x_j}\right] u_i^* dx\, dt.$$

Applying integration by parts with $u = [(I-\alpha^2\Delta)^{-1}u_i^*]$ and $v' = \frac{\partial}{\partial x_j}(u'_j u_i)$

$$\int_0^T \int_{\mathbb{T}^3} \left[(I-\alpha^2\Delta)^{-1}u' \cdot \nabla u\right] \cdot u^* dx\, dt = -\int_0^T \int_{\mathbb{T}^3} u'_j u_i \frac{\partial}{\partial x_j}\left[(I-\alpha^2\Delta)^{-1}u_i^*\right] dx\, dt.$$

Because the goal is to extract u' and $v \cdot w = v^T \cdot w^T$, we can apply a transpose to the above equation and so swap the indices to,

$$\int_0^T \int_{\mathbb{T}^3} \left[(I-\alpha^2\Delta)^{-1}u' \cdot \nabla u\right] \cdot u^* dx\, dt = -\int_0^T \int_{\mathbb{T}^3} u'_i u_j \frac{\partial}{\partial x_i}\left[(I-\alpha^2\Delta)^{-1}u_j^*\right] dx\, dt.$$

Converting back to vector notation,

$$\int_0^T \int_{\mathbb{T}^3} \left[(I-\alpha^2\Delta)^{-1}u' \cdot \nabla u\right] \cdot u^* dx\, dt = -\int_0^T \int_{\mathbb{T}^3} u' \cdot \left[u \cdot \left[\nabla(I-\alpha^2\Delta)^{-1}u^*\right]^T\right] dx\, dt.$$

Starting with the fourth part of the integral and expressing it in Einstein Summation notation,

$$\int_0^T \int_{\mathbb{T}^3} \nabla p' \cdot u^* dx\, dt = \int_0^T \int_{\mathbb{T}^3} \frac{\partial p'}{\partial x_i} u_i^* dx\, dt.$$

Applying integration by parts with $u = u_i^*$ and $v' = \partial x_i p'$,

$$\int_0^T \int_{\mathbb{T}^3} \nabla p' \cdot u^* dx\, dt = -\int_0^T \int_{\mathbb{T}^3} p' \frac{\partial u_i^*}{\partial x_i} dx\, dt.$$

Converting back to vector notation,

$$\int_0^T \int_{\mathbb{T}^3} \nabla p' \cdot u^* dx\, dt = -\int_0^T \int_{\mathbb{T}^3} p' (\nabla \cdot u^*) dx\, dt.$$

Starting with the final part of the integral and expressing it in Einstein Summation notation,

$$\int_0^T \int_{\mathbb{T}^3} (\nabla \cdot u') p^* dx\, dt = \int_0^T \int_{\mathbb{T}^3} \frac{\partial u_i'}{\partial x_i} p^* dx\, dt.$$

Applying integration by parts with $u = p^*$ and $v' = \partial x_i u_i'$,

$$\int_0^T \int_{\mathbb{T}^3} (\nabla \cdot u') p^* dx\, dt = -\int_0^T \int_{\mathbb{T}^3} \frac{\partial p^*}{\partial x_i} u_i' dx\, dt.$$

Converting back to vector notation,

$$\int_0^T \int_{\mathbb{T}^3} (\nabla \cdot u') p^* dx\, dt = -\int_0^T \int_{\mathbb{T}^3} u' \cdot \nabla p^* dx\, dt.$$

Combining the integrals,

$$\int_{\mathbb{T}^3} u^* u' \big|_{t=0}^{t=T} dx - \int_0^T \int_{\mathbb{T}^3} u' \left[\partial_t u^* - \left[u \cdot \nabla \left[(I - \alpha^2 \Delta)^{-1} u^* \right] \right] - \left[u \cdot \left[\nabla (I - \alpha^2 \Delta)^{-1} u^* \right]^T \right] - \nabla p^* \right] - p' (\nabla \cdot u^*)$$

This gives us our adjoint system,

$$\mathbb{L}^* \begin{bmatrix} u_\alpha^* \\ p_\alpha^* \end{bmatrix} := \begin{bmatrix} -\partial_t u^* - (\nabla(I-\alpha^2)^{-1} u_\alpha^* + (\nabla(I-\alpha^2)^{-1} u_\alpha^*)^T) u_\alpha - \nabla p_\alpha^* \\ -\nabla \cdot u_\alpha^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

To derive the terminal condition of the adjoint system we compare the second term of the duality pairing to the objective functional.

$$\int_{\mathbb{T}^3} u'(x,T) \boldsymbol{u}^*(x,T) dx = 2 \left\langle u(x,T), u'(x,T) \right\rangle_{\dot{H}^1}$$

$$= 2 \int_{\mathbb{T}^3} |D| u(x,T) \boldsymbol{u}'(x,T) dx$$

$$= \int_{\mathbb{T}^3} 2|D|^2 u(x,T) \boldsymbol{u}'(x,T) dx$$

Therefore,

$$\boldsymbol{u}^*(x,T) = 2|D|^2 u(x,T)$$

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